The Horizon

Rev: 1.0

Flight Controller on STM32H7 Series

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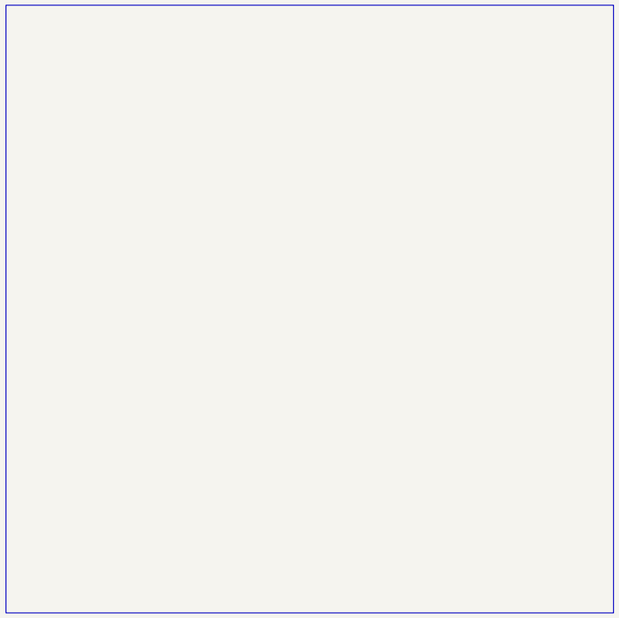
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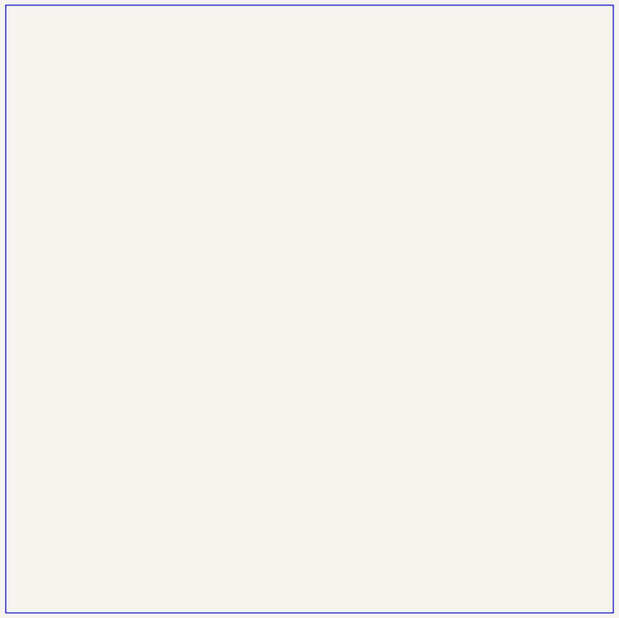
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Top View



Bottom View



Legend

DESIGN NOTE:
Text for information design.

DEBUG NOTE:
Text for Debug notes.

DESIGN NOTE:
Text for design guideline

DESIGN NOTE:
Text for critical design

LAYOUT NOTE:
Text for layout guideline

LAYOUT NOTE:
Text for Critical layout

General Note

Warning Note

Text to be added to silkscreen.

Project: AeroOptix - Horizon			NP-AO-2601-V1-0
Title: **Project Overview**	Drawing by:	Date:	Variant:
	Reviewed by:	Date:	Rev: 1.0 / Date:
KiCad E.D.A. 10.0.8	Sheet: / File: AO-Horizon.kicad_sch	Size: A3	Sheet: A of 4

[B] Block Diagram

Project: AeroOptix - Horizon			NP-AO-2601-V1-0
Title: **Block Diagram**	Drawing by:	Date:	Variant:
	Reviewed by:	Date:	Rev: 1.0 / Date:
KiCad E.D.A. 10.0.8	Sheet: /Block Diagram/ File: block_diagram.kicad_sch	Size: A3	Sheet: B of 4

[C] Top Hierarchical

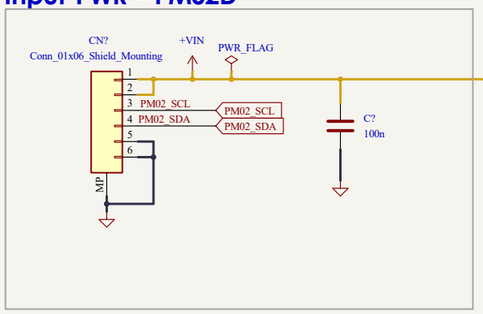
Power Supplies

File: power_supplies.kicad_sch

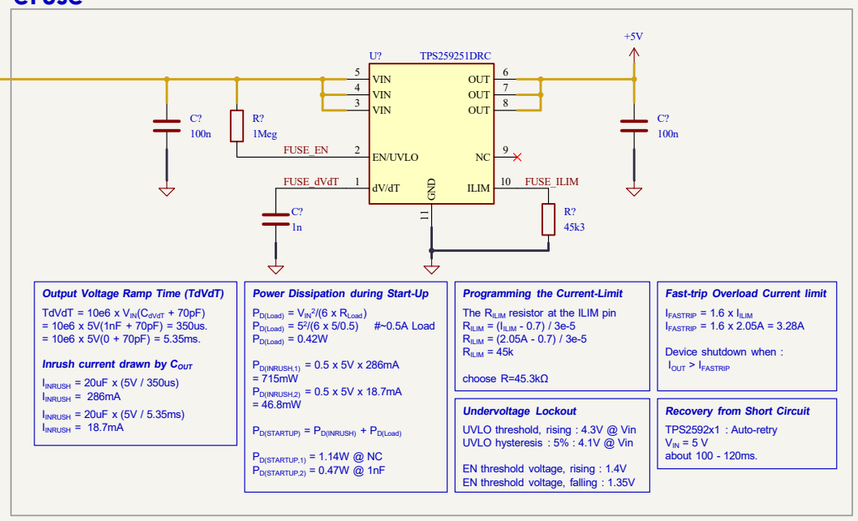
Project: AeroOptix - Horizon			NP-AO-2601-V1-0
Title: **Top Hierarchical**	Drawing by:	Date:	Variant:
	Reviewed by:	Date:	Rev: 1.0 / Date:
KiCad E.D.A. 10.0.8	Sheet: /Top/ File: top_hierarchical.kicad_sch	Size: A3	Sheet: C of 4

[1] Power Supplies

Input PWR - PM02D



eFuse



Output Voltage Ramp Time (TdVdT)

TdVdT = 10e6 x V_{IN}(C_{dVdT} + 70pF)
= 10e6 x 5V(1nF + 70pF) = 350us.
= 10e6 x 5V(0 + 70pF) = 5.35ms.

Inrush current drawn by C_{OUT}

I_{INRUSH} = 20uF x (5V / 350us)
I_{INRUSH} = 286mA

I_{INRUSH} = 20uF x (5V / 5.35ms)
I_{INRUSH} = 18.7mA

Power Dissipation during Start-Up

$P_{D(Load)}$ = V_{IN}^2/(6 x R_{Load})
$P_{D(Load)}$ = 5^2/(6 x 5/0.5) #~0.5A Load
$P_{D(Load)}$ = 0.42W

$P_{D(INRUSH,1)}$ = 0.5 x 5V x 286mA
= 715mW
$P_{D(INRUSH,2)}$ = 0.5 x 5V x 18.7mA
= 46.8mW

$P_{D(STARTUP)}$ = $P_{D(INRUSH)}$ + $P_{D(Load)}$

$P_{D(STARTUP,1)}$ = 1.14W @ NC
$P_{D(STARTUP,2)}$ = 0.47W @ 1nF

Programming the Current-Limit

The R_{ILIM} resistor at the ILIM pin
R_{ILIM} = (I_{ILIM} - 0.7) / 3e-5
R_{ILIM} = (2.05A - 0.7) / 3e-5
R_{ILIM} = 45k

choose R=45.3kΩ

Undervoltage Lockout

UVLO threshold, rising : 4.3V @ Vin
UVLO hysteresis : 5% : 4.1V @ Vin

EN threshold voltage, rising : 1.4V
EN threshold voltage, falling : 1.35V

Fast-trip Overload Current limit

$I_{FASTRIP}$ = 1.6 x I_{ILIM}
$I_{FASTRIP}$ = 1.6 x 2.05A = 3.28A

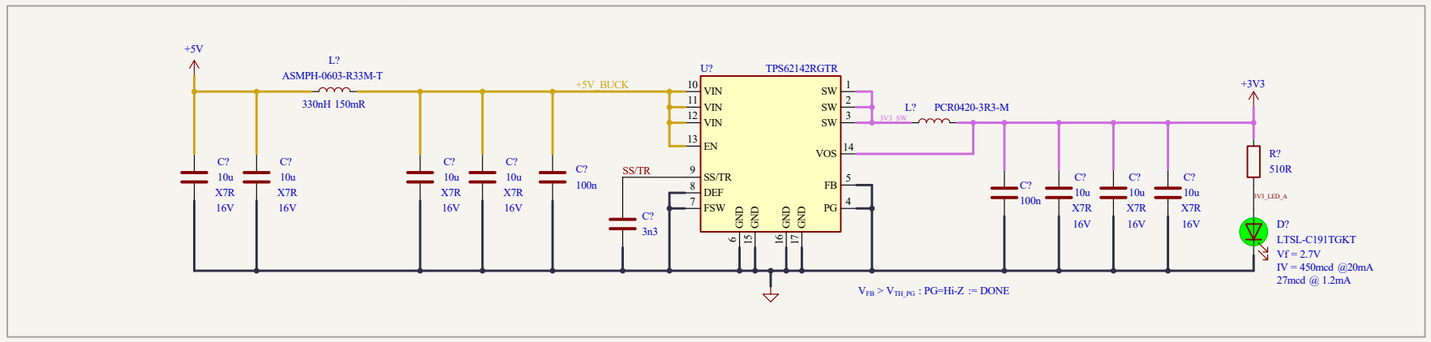
Device shutdown when :
I_{OUT} > $I_{FASTRIP}$

Recovery from Short Circuit

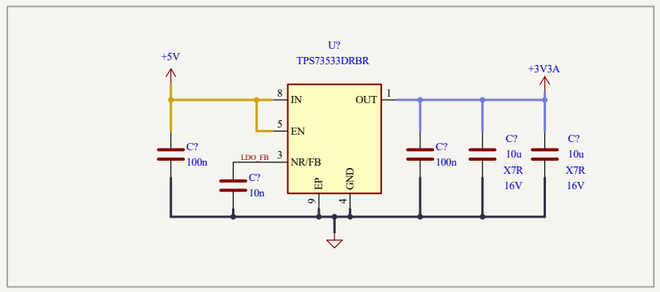
TPS2592x1 : Auto-retry
V_{IN} = 5 V
about 100 - 120ms.

Code	Value	Code	Value	Code	Value	Code	Value	Code	Value	Code	Value
01	100	17	147	33	215	49	316	65	464	81	681
02	102	18	150	34	221	50	324	66	475	82	698
03	105	19	154	35	226	51	332	67	487	83	715
04	107	20	158	36	232	52	340	68	499	84	732
05	110	21	162	37	237	53	348	69	511	85	750
06	113	22	165	38	243	54	357	70	523	86	768
07	115	23	169	39	249	55	365	71	536	87	787
08	118	24	174	40	255	56	374	72	549	88	806
09	121	25	178	41	261	57	383	73	562	89	825
10	124	26	182	42	267	58	392	74	576	90	845
11	127	27	187	43	274	59	402	75	590	91	866
12	130	28	191	44	280	60	412	76	604	92	887
13	133	29	196	45	287	61	422	77	619	93	909
14	137	30	200	46	294	62	432	78	634	94	931
15	140	31	205	47	301	63	442	79	649	95	953
16	143	32	210	48	309	64	453	80	665	96	976

Buck Converter 3V3 2A (2.5MHz)



LDO 3V3 500mA



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Title: **Power Supplies**	Drawing by:	Date:	Variant:
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KiCad E.D.A. 10.0.8	Sheet: /Top/Power Supplies/ File: power_supplies.kicad_sch	Size: A3	Sheet: 1 of 4

[E] Revision History

Project: AeroOptix - Horizon			NP-AO-2601-V1-0	
Title: **Revision History**	Drawing by:	Date:	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date:
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